

Chapter 6C: Thyroid Disease

Hyperthyroidism, Hypothyroidism, Thyroiditis, Nodules, and Thyroid Cancer

This chapter is an original Medvale study resource. The uploaded endocrine thyroid section was used as a topic-coverage signal, including the source emphasis on Graves disease, toxic nodular disease, thyroid storm, hypothyroidism, thyroiditis, nodules, and thyroid cancers. Clinical thresholds and management pathways are cross-checked against thyroid society guidance, but bedside care should follow local protocols and specialist recommendations.

Core goal: turn thyroid lab patterns into illness scripts. Start with TSH, interpret free hormone values, decide whether the gland is overproducing hormone, leaking stored hormone, underproducing hormone, or harboring a structural nodule, then match treatment to mechanism.

Section	Clinical task
Thyroid physiology	Use negative feedback to explain TSH/T4/T3 patterns.
Thyrotoxicosis	Separate Graves/toxic nodules from thyroiditis and exogenous hormone.
Hypothyroidism	Recognize primary vs central disease and treat safely.
Thyroiditis	Use pain, uptake, antibodies, and phase pattern.
Nodules/cancer	Risk-stratify nodules and identify cancer phenotypes.
Emergencies	Treat thyroid storm and myxedema coma before confirmation when unstable.

1. Why this matters clinically

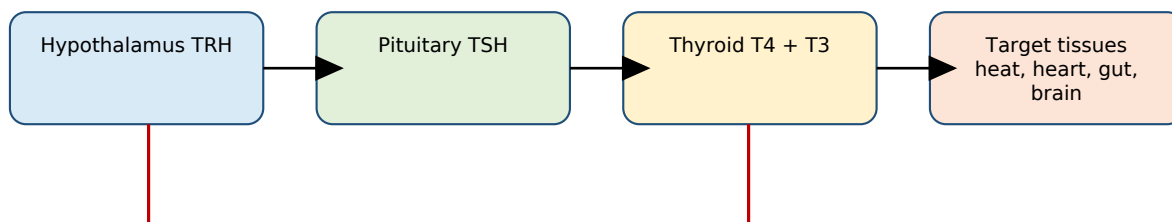
Thyroid disease is common because the hypothalamic-pituitary-thyroid axis is sensitive, measurable, and easily perturbed by autoimmunity, iodine exposure, medications, pregnancy, severe illness, and structural thyroid disease. A small change in circulating thyroid hormone can produce large downstream effects on heart rate, vascular tone, heat production, bowel motility, neuromuscular excitability, mood, menstrual function, lipid metabolism, and bone turnover. For internal medicine, the highest-yield skill is not memorizing every subtype; it is recognizing when the laboratory pattern and the physical findings point to increased synthesis, destructive release, impaired production, or malignancy risk.

A thyroid diagnosis should be anchored to mechanism. Graves disease stimulates the TSH receptor. Toxic multinodular goiter and toxic adenoma autonomously synthesize hormone. Thyroiditis releases preformed hormone from damaged follicles, so uptake is low and antithyroid synthesis blockers do not fix the leak. Primary hypothyroidism reflects gland failure and produces a high TSH; central hypothyroidism reflects inadequate pituitary or hypothalamic drive and produces low or inappropriately normal TSH with low free T4. Nodules require a different logic: determine functional status and sonographic cancer risk before deciding on surveillance, FNA, radionuclide imaging, or surgery.

Numerical anchors

Anchor	Interpretation
TSH	Most sensitive screening test for primary thyroid dysfunction; suppressed in most overt thyrotoxicosis and elevated in primary hypothyroidism.
Free T4	Defines overt disease when abnormal; use with TSH to detect central disease and severe illness patterns.
T3	Can be disproportionately elevated in early Graves or toxic nodular disease; consider when TSH is low and free T4 is normal but symptoms persist.
Radioiodine uptake	High in increased hormone synthesis; low in thyroiditis, exogenous thyroid hormone, or iodine load.
Thyroid cancer in evaluated nodules	Most nodules are benign; malignancy risk is driven by ultrasound pattern, size, history, and cytology, not palpation alone.
Thyroid storm mortality	Life-threatening; requires multi-drug treatment and ICU-level monitoring.
Myxedema coma	Endocrine emergency with altered mental status, hypothermia, hypoventilation, and high mortality even with treatment.

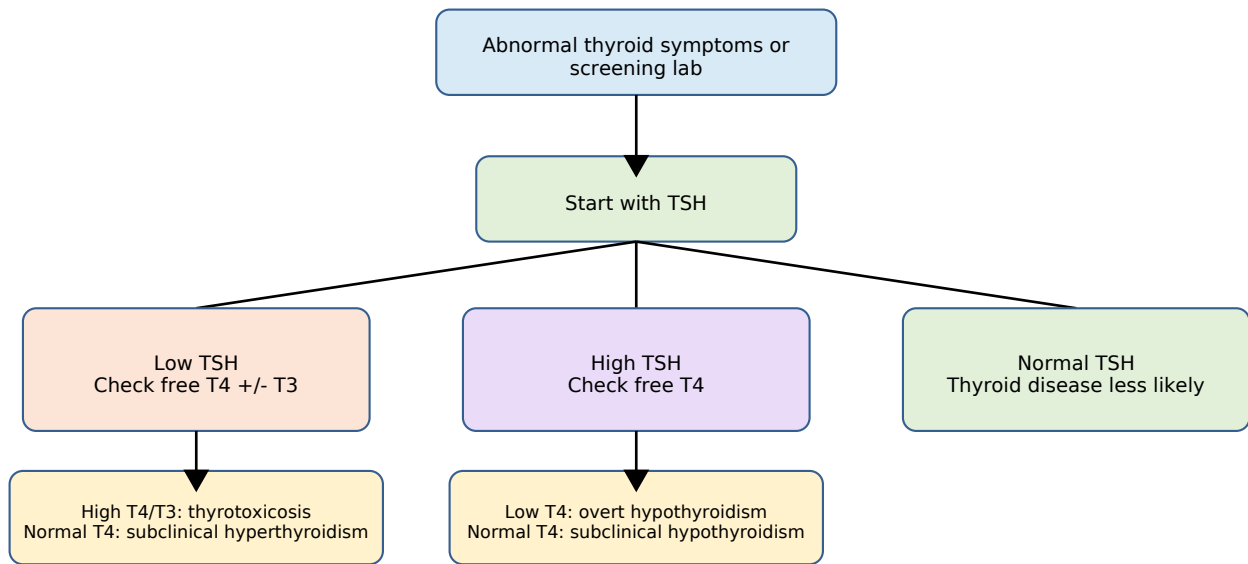
2. Thyroid axis physiology and lab interpretation



Rising free T4/T3 suppresses TRH and TSH; falling hormone releases the brake.

The axis behaves like a thermostat. TRH from the hypothalamus stimulates pituitary TSH; TSH stimulates thyroid iodide uptake, thyroglobulin iodination, coupling reactions, and release of mostly T4 with smaller amounts of T3. Peripheral deiodinases convert T4 into active T3 or inactive reverse T3. Free hormone feeds back on the hypothalamus and pituitary. Because TSH responds logarithmically to free hormone, mild thyroid hormone changes can cause large TSH shifts.

Most circulating T4 and T3 is protein-bound, especially to thyroxine-binding globulin. Total hormone values change when binding proteins change, such as pregnancy, estrogen therapy, liver disease, or nephrotic-range protein loss. Free T4 is usually preferred when binding abnormalities are suspected. Non-thyroidal illness can lower T3 and sometimes TSH transiently, so thyroid testing during severe illness should be targeted rather than reflexive.



Common pitfall

Do not diagnose hyperthyroidism from symptoms alone. Anxiety, weight loss, tremor, palpitations, diarrhea, atrial fibrillation, and insomnia have broad differentials. Conversely, older adults may have apathetic hyperthyroidism with weight loss, depression, weakness, or atrial fibrillation rather than classic heat intolerance and hyperactivity.

3. Hyperthyroidism and thyrotoxicosis

Thyrotoxicosis means excess thyroid hormone effect at tissues. Hyperthyroidism is the subset caused by increased thyroid hormone synthesis. This distinction matters because thionamides block new hormone synthesis but do not immediately stop the release of already formed hormone from an inflamed gland. The first laboratory pattern is usually low TSH; overt disease has high free T4 and/or high T3, while subclinical hyperthyroidism has low TSH with normal free hormone levels.

Illness scripts

Condition	Mechanism	Key clues	Uptake pattern	Treatment logic
Graves disease	TSH receptor-stimulating antibodies	Diffuse goiter, bruit, ophthalmopathy, pretibial myxedema, other autoimmune disease	Diffuse high uptake	Beta-blocker for symptoms; methimazole for most nonpregnant adults; radioiodine or surgery for selected patients.
Toxic multinodular goiter	Autonomous nodules	Older patient, irregular nodular thyroid, gradual thyrotoxicosis	Patchy/focal high uptake	Radioiodine, surgery, or medical bridge depending on size, symptoms, and comorbidity.
Toxic adenoma	Single autonomous nodule	Solitary hot nodule; surrounding gland suppressed	Focal high uptake	Radioiodine or surgery; malignancy uncommon in hot nodules.
Subacute painful thyroiditis	Post-viral destructive release	Tender thyroid, fever or URI prodrome, high ESR/CRP	Low uptake	NSAIDs or glucocorticoids for pain; beta-blocker if needed.

Painless/postpartum thyroiditis	Autoimmune destructive release	No thyroid pain; may cycle hyper to hypo to recovery	Low uptake	Supportive; beta-blocker for symptoms; monitor for hypothyroid phase.
Exogenous thyroid hormone	Ingestion or overreplacement	Low thyroglobulin if factitious; no goiter unless underlying disease	Low uptake	Reduce or stop hormone; supportive beta-blocker.
Iodine-induced thyrotoxicosis	Jod-Basedow in autonomous tissue or amiodarone-related	Iodinated contrast, amiodarone, supplements	Variable	Mechanism-specific; often endocrine consultation.

Classic symptoms include nervousness, insomnia, heat intolerance, sweating, weight loss despite appetite, frequent stools, muscle weakness, tremor, palpitations, menstrual irregularity, and sometimes dyspnea from high-output physiology or atrial fibrillation. Signs include tachycardia, widened pulse pressure, warm moist skin, tremor, hyperreflexia, proximal myopathy, and goiter. Graves-specific findings include orbitopathy, pretibial myxedema, thyroid bruit, and diffuse nontender thyroid enlargement.

Diagnostic approach

Confirm the pattern with TSH and free T4; add total or free T3 when T3-toxicosis is suspected. Determine mechanism with TSH receptor antibodies when Graves is likely, radionuclide uptake when the cause is unclear and pregnancy is excluded, and ultrasound when structural nodules must be characterized. A low uptake state should redirect the clinician away from antithyroid synthesis drugs as primary therapy and toward thyroiditis, exogenous hormone, or iodine exposure.

Management

Beta-blockers provide rapid symptomatic control, especially for tremor, palpitations, and tachyarrhythmias. Methimazole is preferred for most nonpregnant adults because it is effective and generally easier to dose than PTU. PTU is used in thyroid storm and is often preferred in the first trimester of pregnancy because methimazole has teratogenic concerns early in gestation. Both thionamides can cause rash, arthralgias, hepatotoxicity, and agranulocytosis; fever or sore throat while taking these drugs should trigger urgent evaluation. Radioiodine ablation and thyroidectomy are definitive options; pregnancy and breastfeeding are contraindications to radioiodine.

Thyroid storm

Thyroid storm is not just severe lab hyperthyroidism; it is decompensated thyrotoxicosis with fever, marked tachycardia or atrial fibrillation, heart failure, agitation, delirium, psychosis, vomiting, diarrhea, hepatic injury, or shock. Triggers include infection, surgery, trauma, myocardial infarction, DKA, parturition, and antithyroid drug interruption. Treat immediately: supportive care and cooling, IV fluids and glucose as needed, beta-blockade, PTU or methimazole, iodine at least 1 hour after thionamide, glucocorticoids, and treatment of the precipitant.

4. Hypothyroidism and myxedema coma

Primary hypothyroidism is thyroid gland failure and accounts for most cases. Hashimoto thyroiditis is the typical autoimmune cause in iodine-sufficient settings, while prior radioiodine therapy, thyroidectomy, neck radiation, iodine deficiency, lithium, amiodarone, immune checkpoint inhibitors, and infiltrative disease are other important causes. Secondary or tertiary hypothyroidism is central disease; TSH can be low, normal, or mildly elevated but biologically inadequate, so free T4 is essential.

Pattern	Likely diagnosis	Typical next step
High TSH + low free T4	Overt primary hypothyroidism	Levothyroxine; check etiology and comorbid risk.
High TSH + normal free T4	Subclinical hypothyroidism	Repeat testing, thyroid peroxidase antibodies, treat if TSH persistently high, symptoms, pregnancy/fertility, or high-risk context.
Low/inappropriately normal TSH + low free T4	Central hypothyroidism	Evaluate pituitary axis; treat adrenal insufficiency before thyroid replacement.

Low T3 with variable TSH/T4 in severe illness	Non-thyroidal illness pattern	Avoid reflex thyroid hormone unless true hypothyroidism is established.
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Symptoms often accumulate slowly: fatigue, cold intolerance, constipation, dry skin, coarse hair, hoarseness, weight gain that is usually modest, menstrual irregularity, depression, cognitive slowing, muscle cramps, arthralgias, and decreased hearing. Signs include bradycardia, delayed relaxation of deep tendon reflexes, periorbital puffiness, nonpitting edema from glycosaminoglycan deposition, carpal tunnel syndrome, macroglossia, and a firm rubbery goiter in Hashimoto disease. Labs may show hyponatremia, hyperlipidemia, elevated CK, mild anemia, or hyperprolactinemia.

Treatment is levothyroxine, usually once daily on an empty stomach. In older adults or patients with coronary artery disease, start low and titrate because thyroid hormone increases myocardial oxygen demand. Recheck TSH after dose changes, usually after about 6 weeks, because TSH equilibrates slowly. In central hypothyroidism, dose by free T4 rather than TSH. Always consider adrenal insufficiency before thyroid hormone in suspected pituitary disease or myxedema coma.

Myxedema coma

Myxedema coma is severe hypothyroidism with altered mental status, hypothermia, hypoventilation, bradycardia, hypotension, hyponatremia, hypoglycemia, and often a precipitating infection, cold exposure, sedative, trauma, stroke, or myocardial event. Treat empirically with ICU care, passive rewarming, ventilatory support if needed, stress-dose glucocorticoids, IV levothyroxine, and management of the precipitant.

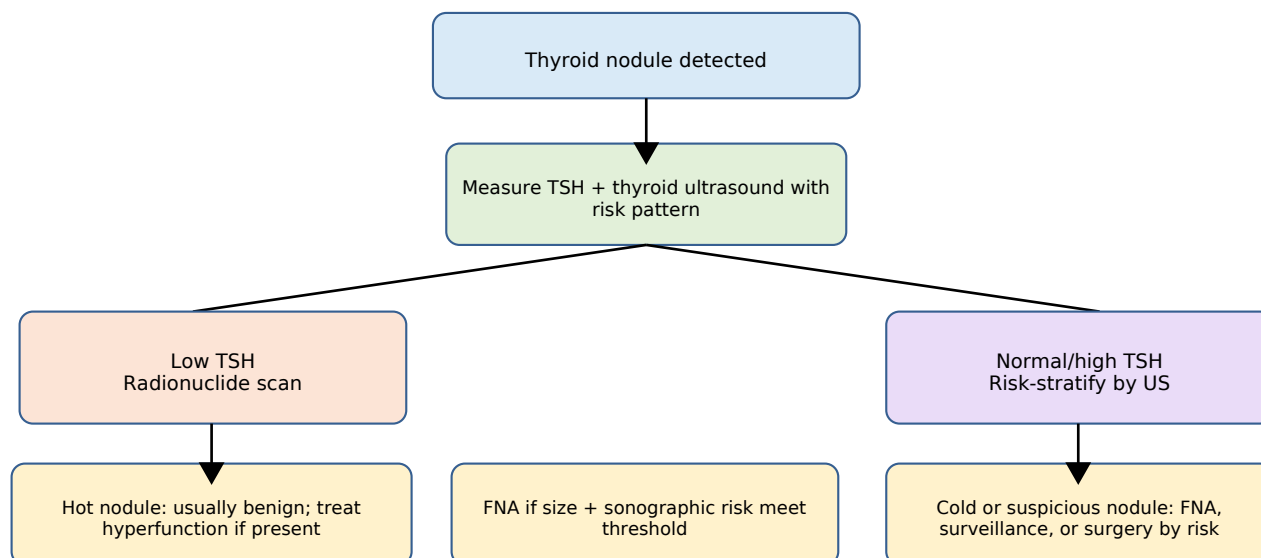
5. Thyroiditis

Thyroiditis disorders are best understood as inflammatory or destructive processes that may produce a triphasic course: transient thyrotoxicosis from leakage of stored hormone, followed by transient hypothyroidism after stores are depleted, followed by recovery or permanent hypothyroidism. Uptake is low during destructive thyrotoxicosis because the gland is not actively synthesizing new hormone.

Type	Pain	Mechanism/context	Labs/imaging	Treatment
Subacute granulomatous/de Quervain	Painful/tender	Often post-viral; fever, neck pain radiating to jaw/ear	Low TSH early, high ESR/CRP, low uptake	NSAIDs; glucocorticoids if severe; beta-blocker for symptoms.
Painless/silent	Painless	Autoimmune; may recur	Low uptake, antibodies may be present	Supportive; monitor for hypothyroidism.
Postpartum	Painless	Within 1 year after delivery	Low uptake; breastfeeding affects nuclear imaging decisions	Beta-blocker if symptomatic; levothyroxine during symptomatic hypo phase or pregnancy planning.
Hashimoto	Usually painless	Chronic lymphocytic autoimmune gland destruction	TPO antibodies common; may have goiter	Levothyroxine when hypothyroid.
Riedel fibrosing	Firm fixed gland	Fibrosis replacing thyroid; may mimic cancer	Hard thyroid, compressive features	Specialist evaluation; surgery for compression; hormone if hypothyroid.
Suppurative	Painful	Bacterial infection, immunocompromise, anatomic fistula	Fever, focal abscess	Antibiotics and drainage.

6. Thyroid nodules

Thyroid nodules are common, especially with age and increased imaging. The key task is to avoid both underdiagnosing clinically important cancers and overtreating benign or indolent nodules. History should assess childhood head/neck radiation, family history of thyroid cancer or MEN2, rapid growth, hoarseness, dysphagia, dyspnea, prior thyroid disease, and compressive symptoms. Physical exam should look for fixation to surrounding tissue, cervical lymphadenopathy, vocal cord dysfunction, and a hard solitary nodule.



Ultrasound is central because cancer risk is more closely tied to sonographic pattern than to palpation alone. Suspicious patterns include solid hypoechoic composition, irregular or infiltrative margins, microcalcifications, taller-than-wide shape, extrathyroidal extension, and abnormal cervical lymph nodes. Purely cystic nodules are rarely malignant, but mixed or solid nodules still need risk-based evaluation. FNA is used when size and ultrasound pattern meet threshold; indeterminate cytology may lead to repeat FNA, molecular testing, diagnostic lobectomy, or surveillance depending on risk and patient preference.

Hot vs cold logic

A suppressed TSH suggests autonomy. A radionuclide scan can identify a hyperfunctioning 'hot' nodule, which is rarely malignant and is usually managed as a cause of hyperthyroidism rather than by FNA. A nonfunctioning 'cold' nodule is not automatically cancer, but if the ultrasound risk pattern and size warrant sampling, FNA is appropriate.

7. Thyroid cancer

Most thyroid cancers are differentiated cancers arising from follicular cells. Prognosis depends on histology, age, size, extrathyroidal extension, nodal disease, distant metastasis, and molecular features. A calm approach is important: many papillary cancers are indolent, while anaplastic cancer is an airway-threatening emergency.

Cancer	Typical features	Spread	Markers/clues	Management frame
Papillary	Most common; associated with radiation; Orphan Annie nuclei, grooves, psammoma bodies	Lymphatic to cervical nodes	Often positive iodine uptake; excellent prognosis	Surgery based on risk/extent; possible radioiodine and TSH suppression.
Follicular	Follicular cells with capsular or vascular invasion; may be iodine avid	Hematogenous to bone, lung, liver	Cannot diagnose by FNA alone because invasion is required	Lobectomy for diagnosis or thyroidectomy by risk; radioiodine if indicated.
Hurthle cell	Oncocytic variant; less iodine avid	Local/distant	More aggressive than typical follicular	Surgical management; radioiodine less reliable.
Medullary	Parafollicular C-cell tumor	Nodes and distant sites	Calcitonin, CEA; MEN2/RET; screen for pheochromocytoma before surgery	Total thyroidectomy with nodal management; radioiodine not useful.

Anaplastic	Elderly, rapidly enlarging neck mass, hoarseness, dysphagia, airway compromise	Highly invasive	Very poor prognosis	Urgent airway, oncology, surgery/radiation/systemic therapy; palliative goals often needed.
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Medullary carcinoma deserves separate recognition because it arises from C cells and produces calcitonin. A patient with suspected medullary carcinoma should be evaluated for hereditary disease, especially MEN2, and pheochromocytoma must be excluded before thyroid surgery to avoid catecholamine crisis.

8. Integrated diagnostic approach

Presentation	First pattern to check	High-yield next branch
Palpitations, tremor, weight loss	TSH low? Free T4/T3 high?	Graves antibodies or uptake scan; beta-blocker while evaluating if symptomatic.
Atrial fibrillation in older adult	TSH	Do not miss apathetic hyperthyroidism; treat arrhythmia and thyroid driver.
Fatigue, cold intolerance, constipation	TSH + free T4	Primary vs central; assess lipids, anemia, sodium; start levothyroxine thoughtfully.
Painful anterior neck after viral illness	TSH/free T4 + ESR/CRP	Low uptake thyroiditis; NSAIDs/steroids rather than thionamides.
Thyroid nodule	TSH + ultrasound	Low TSH -> scan; normal/high TSH -> US risk and FNA threshold.
Fever, delirium, tachycardia, diarrhea in thyrotoxic patient	Do not wait for confirmatory tests	Treat thyroid storm immediately.
Hypothermia, coma, bradycardia, hyponatremia	Draw TSH/free T4/cortisol but treat	Stress-dose steroids then IV thyroid hormone and ICU support.

9. Common pitfalls

- Treating thyroiditis thyrotoxicosis with methimazole as if it were Graves disease. Low uptake destructive thyroiditis needs symptom control and anti-inflammatory care, not synthesis blockade.
- Forgetting pregnancy before radioiodine uptake testing or ablation. Pregnancy and breastfeeding change diagnostic and treatment choices.
- Using total T4 without considering binding protein changes. Pregnancy and estrogen increase TBG and can raise total hormone without true hyperthyroidism.
- Relying on TSH alone in suspected central hypothyroidism. Central disease can have a deceptively normal TSH.
- Starting levothyroxine aggressively in an older patient with coronary disease. Rapid replacement can provoke angina or arrhythmia.
- Giving iodine before thionamide in thyroid storm. Iodine should follow synthesis blockade to avoid fueling new hormone production.
- Assuming a cold nodule is cancer or a cystic nodule is never relevant. Risk depends on the whole ultrasound and clinical context.
- Missing medullary thyroid cancer clues: calcitonin elevation, MEN2, family history, and the need to exclude pheochromocytoma before surgery.

10. Practice questions

Question 1. A 32-year-old woman has weight loss, tremor, diffuse goiter, and diplopia. TSH is suppressed and free T4 is elevated. What single finding most specifically supports Graves disease?

Answer: Ophthalmopathy from extraocular muscle and retro-orbital tissue inflammation is highly suggestive of Graves disease. Pretibial myxedema and a thyroid bruit are also Graves-specific clues.

Question 2. A 45-year-old man has neck pain after a viral illness, fever, low TSH, high free T4, and very low radioactive iodine uptake. What is the preferred treatment mechanism?

Answer: This is subacute thyroiditis with destructive release of stored hormone. Treat pain/inflammation with NSAIDs or glucocorticoids and adrenergic symptoms with a beta-blocker; thionamides are not the primary therapy.

Question 3. A 76-year-old woman presents with hypothermia, bradycardia, confusion, hypoventilation, and hyponatremia. TSH/free T4 are pending. What should be given before or with thyroid hormone?

Answer: Stress-dose glucocorticoids should be given because adrenal insufficiency may coexist or be unmasked. She needs ICU-level support and IV thyroid hormone.

Question 4. A patient with a 2-cm thyroid nodule has low TSH. What is the next test before FNA?

Answer: Radionuclide scan. A hyperfunctioning hot nodule is rarely malignant and is usually managed as autonomous thyroid hormone production rather than sampled first.

Question 5. A thyroid FNA suggests follicular neoplasm. Why can FNA be insufficient for definitive follicular carcinoma diagnosis?

Answer: Follicular carcinoma requires capsular or vascular invasion, which cytology alone cannot reliably establish. Diagnostic lobectomy may be needed.

Question 6. A patient with medullary thyroid carcinoma is scheduled for thyroidectomy. What endocrine tumor must be excluded first?

Answer: Pheochromocytoma, especially in MEN2, because surgery without alpha-blockade can trigger catecholamine crisis.

11. Graph-RAG extraction map

Node	Class	Key attributes	High-value edges
TSH	Diagnostic lab	Primary screening marker; suppressed in hyperthyroidism; elevated in primary hypothyroidism	Regulated by free T4/T3; orders free T4 interpretation; altered in pituitary disease
Graves disease	Autoimmune endocrine disease	TSH receptor antibodies; diffuse goiter; ophthalmopathy; pretibial myxedema	Causes hyperthyroidism; treated by beta-blockers, methimazole, radioiodine, surgery
Thyroiditis	Inflammatory endocrine disease	Low uptake thyrotoxicosis; painful or painless; triphasic course	Mimics Graves; causes transient hyperthyroidism and hypothyroidism
Hashimoto thyroiditis	Autoimmune endocrine disease	TPO antibodies; firm goiter; primary hypothyroidism	Causes high TSH; treated with levothyroxine; associated with other autoimmune disorders
Thyroid nodule	Structural thyroid lesion	Risk assessed by TSH, ultrasound, FNA	Can be hot/cold; may represent adenoma or carcinoma
Papillary thyroid carcinoma	Differentiated thyroid cancer	Most common; lymphatic spread; radiation risk	Managed by surgery +/- radioiodine/TSH suppression
Medullary thyroid carcinoma	C-cell cancer	Calcitonin; MEN2/RET; pheochromocytoma screening	Requires total thyroidectomy; radioiodine ineffective
Thyroid storm	Endocrine emergency	Fever, tachyarrhythmia, CNS/GI dysfunction	Complication of thyrotoxicosis; treated by beta-blocker, thionamide, iodine, steroids
Myxedema coma	Endocrine emergency	Hypothermia, AMS, bradycardia, hypoventilation, hyponatremia	Complication of hypothyroidism; treated by hydrocortisone, IV levothyroxine, ICU support

Suggested edge types for the graph: causes, mimics, diagnosed-by, treated-by, contraindicated-in, progresses-to, associated-with, emergency-management, and complication-of. A useful retrieval pattern is: symptom cluster -> TSH/free T4 pattern -> mechanism class -> treatment class -> safety exception.

References and guideline cross-checks

- American Thyroid Association. Clinical Practice Guidelines and Statements: hyperthyroidism, hypothyroidism, thyroid nodules, thyroid cancer, thyroid eye disease, and anaplastic thyroid cancer guidance pages.

- Ross DS, Burch HB, Cooper DS, et al. 2016 American Thyroid Association Guidelines for Diagnosis and Management of Hyperthyroidism and Other Causes of Thyrotoxicosis.
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